

Green Industrial Engineering on the Web: Agent-Driven Ant Colony Optimization Tuning for Energy-Efficient 3D Pipe Routing

Xuanhan Fan*
Beijing Institute of Technology
Beijing, China
Shenzhen MSU-BIT University
Shenzhen, China
fanxuanhan@gmail.com

Rui Ding*
Shenzhen University
Shenzhen, China
ddingrui0127@gmail.com

Haojie Liu*
Shenzhen University
Shenzhen, China
a24130712413@gmail.com

Jibin Zhou†
Dalian Institute of Chemical Physics,
Chinese Academy of Sciences
Dalian, China
zhoujibin@dicp.ac.cn

Han Liu†
Dalian University of Technology
Dalian, China
liu.han.dut@gmail.com

Yuanman Li†
Shenzhen University
Shenzhen, China
yuanmanli@szu.edu.cn

Wei Wang†
Shenzhen MSU-BIT University
Shenzhen, China
ehomewang@ieee.org

Mao Ye
Dalian Institute of Chemical Physics,
Chinese Academy of Sciences
Dalian, China
maoye@dicp.ac.cn

Abstract

With the advancement of networked data, artificial intelligence, and sustainability, industrial engineering is rapidly shifting toward intelligent design and production. Traditional manual design and ant colony optimization (ACO) schemes that rely on manually tuned hyperparameters increasingly exhibit poor generalization, high tuning overhead, and limited ability to simultaneously account for energy efficiency and environmental constraints in complex 3D settings. Focusing on 3D pipeline routing in petrochemical plants, this paper proposes a graph-aware reinforcement learning framework for adaptive ACO hyperparameter tuning, and implements an end-to-end integration and validation on a web-based 3D digital factory platform. Methodologically, we construct a unified 3D grid environment and equipment topology graph, where device geometry and connectivity are encoded as graph features and processed by a multi-head Graph Attention Network (GAT) to capture global spatial-process structure. In parallel, scene scale, obstacle density, connection geometry, vertical complexity, and routing progress are summarized into a compact connection state vector, fused with the graph embedding via cross-attention and passed to a PPO-based Actor-Critic agent that jointly configures ACO hyperparameters as a continuous action. At the algorithmic level, we further introduce a heterogeneous pheromone deposition

rule and a self-regulating pheromone mechanism to form a goal-directed pheromone gradient, coupled with a multi-objective cost function over geometric quality and energy-carbon performance, yielding greener and more adaptive routing than conventional ACO-based schemes. The implementation source can be found at <https://github.com/XuanhanFan/GRL-DEACO>.

CCS Concepts

• **Applied computing** → **Operations research**; • **Computing methodologies** → **Planning for deterministic actions**; • **Information systems** → **Decision support systems**.

Keywords

Green Industrial Engineering, Energy-Efficient Pipe Routing, Ant Colony Optimization, Proximal Policy Optimization

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1 Introduction

Although renewable energy technologies continue to advance, fossil fuels are expected to remain dominant in the global energy mix for the foreseeable future, currently accounting for about 87% of primary energy consumption [12, 50]. Against this backdrop, petrochemical plants, as core infrastructures for processing and transporting fossil resources, pose a tightly coupled spatial design problem that must balance safety, operational and maintenance efficiency, capital cost, and flexibility for future expansion [6]. As

*Both authors contributed equally to this research.

†Corresponding author.



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as a result, pipeline routing is both highly complex and critically important, while current practice still relies heavily on engineers' experience and repeated manual iterations, leading to low design efficiency and elevated costs. Although a wide variety of pipeline routing algorithms are available, they typically rely on numerous, tightly coupled hyperparameters, making it difficult for engineers to efficiently configure them for different connection scenarios and obtain satisfactory routing solutions, as illustrated in Fig. 1.

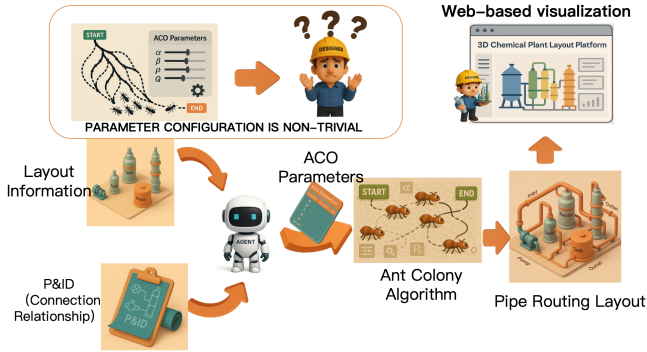


Figure 1: Configuring ACO hyperparameters is non-trivial in complex multi-connection scenes, whereas a reinforcement learning agent can adapt them to each context, reducing design effort and improving path quality.

In 3D routing[2, 19], heuristic deterministic search such as A^* [11, 28] can yield stable, interpretable paths when geometry and constraints are explicitly modeled, but its runtime grows rapidly with state-space dimensionality and constraint complexity. As equipment counts, constraint types (e.g., minimum bend radius, parallel spacing, manufacturability) and degrees of freedom increase, branching factors and node expansions escalate, causing a combinatorial explosion that limits scalability and often prevents feasible, constraint-satisfying solutions from being obtained within practical engineering time.

In recent years, heuristic and metaheuristic methods (e.g., ACO [9, 10], PSO [20], GA [37]) have been widely adopted for complex path planning because their non-determinism helps escape local optima. In industrial piping design, algorithms must satisfy hard constraints such as collision avoidance, clearance and minimum bend radius, while also accommodating soft engineering preferences such as reducing elbows or hugging bulkheads. These soft constraints are typically encoded in intricate heuristic functions whose effectiveness depends heavily on a set of preset weight parameters[30]; in practice, designers must further reselect and retune these weights for different scenes and congestion patterns, which is labor-intensive and brittle.

To address these challenges of manually preset weights and scene-dependent heuristic parameter selection, we propose Graph-aware Reinforcement Learning–tuned DEACO (GRL-DEACO), a reactive intelligent-agent[23, 29] framework for 3D pipeline routing that dynamically adjusts the heuristic parameters of an ant colony optimization (ACO)–based router. GRL-DEACO comprises a state extractor, a graph attention network (GAT) scene encoder [4,

13, 36, 42], a policy–value network [15, 33] augmented with cross-attention [17, 41], an action-mapping and constraint-repair module, an ACO interaction environment with task-specific rewards, and a Proximal Policy Optimization (PPO) trainer [38]. Plant layouts are encoded as attributed graphs so that the agent can jointly capture global topology[16] and local connection state, and adapt heuristic weights and algorithmic control factors to real-time conditions such as obstacle density[51] and path complexity. As illustrated in Fig. 2, GRL-DEACO is further integrated into a web-based[34] design platform, where web technologies provide the front-end interface and platform APIs with JSON-based messages [3] transmit environment configurations and routing requests; during piping design, the agent continuously senses the evolving scene and connection tasks, selects routing hyperparameters and drives the underlying planner to generate feasible, high-quality pipeline paths.

To sum up, our contributions are as follows:

- We develop a physics-informed, energy- and carbon-aware Dual-Strategy Enhanced Ant Colony Optimization (DEACO with green metrics) formulation for 3D petrochemical pipe routing. Geometric path metrics and surrogate energy/ CO_2 models are fused into a unified objective that favors short, manufacturable, and energy-efficient routes.
- We propose Graph-aware Reinforcement Learning–tuned DEACO (GRL-DEACO), a graph-topology–aware reactive agent that represents each plant layout as an attributed scene graph and, via a PPO-trained policy, adaptively tunes ACO heuristic weights and control hyperparameters on a per-connection basis rather than relying on fixed hand-crafted parameters.
- We evaluate GRL-DEACO on petrochemical layouts of varying complexity and find that it consistently produces strictly feasible routes for all target connections and overall outperforms fixed-parameter DEACO. Across the simple, medium, and complex benchmark scenarios, it achieves the lowest average bend count and operating energy. Moreover, on the subset of test connections where its behaviour differs most from DEACO (with green metrics), GRL-DEACO exhibits stronger generalization, yielding routes with fewer bends and lower energy consumption.

2 Related work

Dong et al. [8] proposed a hybrid strategy that combines NSGA-III with multi-population parallel evolution, effectively addressing the difficulty of simultaneously optimizing multiple objectives (such as path length and number of bends) and mitigating the tendency of traditional algorithms to be trapped in local optima in complex 3D ship piping environments. However, multi-objective optimization[46, 47] must still balance feasibility, convergence, and solution diversity in a principled manner. Although Kim et al. [21] introduced a curriculum-learning-based strategy that alleviates convergence issues caused by sparse rewards when applying RL to complex ship environments, their study primarily focuses on static settings and does not fully account for the dynamic evolution of obstacles during sequential pipe routing. Zhang et al. integrated multi-agent RL with a 3D A^* search algorithm in a hierarchical hybrid framework, thereby improving the joint optimization of 3D

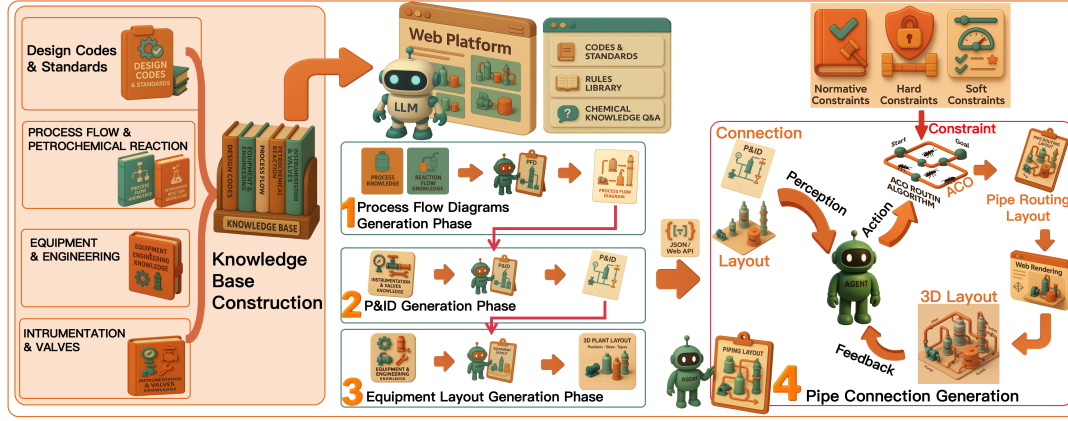


Figure 2: Functional architecture and workflow of the petrochemical agent web platform.

facility layout and piping routes under assembly constraints; nevertheless, the simplified A* component struggles to capture the full spectrum of complex process and code constraints encountered in real engineering practice. Wu et al. [45] employed a multi-objective ant colony system to balance cost and service quality in bus network design. This approach does not transfer directly to chemical piping, and its adaptability to complex operating conditions and flexibility in weight adjustment remain limited. Zhang et al. [49] tackle the coupled optimization of facility layout and pipe routing using a hierarchical hybrid framework that integrates multi-agent reinforcement learning (MARL) with a 3D A* search algorithm. Liu et al. [27] used AI-enhanced ACO to alleviate the blind search problem in 3D pipe layout, and Wang et al. [43] further improved environmental adaptability through a dual-strategy mechanism. Despite clear gains in solution quality, these methods depend on intricate heuristics with many manually tuned parameters and generalize poorly across scenarios. To address this limitation, we introduce a reinforcement-learning-based mechanism [18, 26, 40] that adaptively adjusts key parameters within the feasible design space in response to the current layout and connection task.

3 Proposed method

Problem Definition. At step t , given an attributed scene graph $\mathcal{G}_t = (\mathcal{V}_t, \mathcal{E}_t, \mathbf{X}_t, \mathbf{A}_t)$ and P&ID terminals $C = \{(u, v)\}$ [39], we must route the active pair (u_t, v_t) by finding P_t that is strictly feasible (collision-free with mandated clearances and minimum bend radius) and compliant with topology, manufacturability, and maintenance rules. Accepted routes update the scene to $\mathcal{G}_{t+1}$. We learn a policy $\pi(s_t, \mathcal{G}_t) \mapsto \psi_t$ that parameterizes an ACO solver $\text{ACO}(\psi_t; \mathcal{G}_t, u_t, v_t) \rightarrow P_t$.

3.1 Enhanced ACO Algorithm with Green Metrics

To strike a robust balance between geometric fidelity and computational efficiency, we approximate each placed device mesh by a compact set of axis-aligned *voxel bounding boxes* (VBBs) [31]. The original mesh is voxelized at spacing h , hole-filled, and its connected voxel components are approximated and greedily merged

into axis-aligned boxes under geometric-error and fill-ratio constraints, yielding a box set $\mathcal{B}$ with $|\mathcal{B}| \ll |\mathcal{V}|$. To explicitly encode manufacturability constraints and safety clearances, every box is inflated by an effective clearance radius $r_{\text{eff}} = r_{\text{pipe}} + r_{\text{safe}}$, where r_{pipe} is the physical pipe radius (including insulation, when applicable) and r_{safe} is an additional allowance for installation workspace, tolerances, thermal expansion, maintenance access, and code-mandated separations. The union of all dilated boxes is finally rasterized into a 3D occupancy grid $S \in \{0, 1\}^{M \times N \times L}$ with voxel spacing h over the scene bounds, using the convention that occupied voxels are encoded as 0 and free voxels as 1. This grid serves as the clearance-aware environment for all subsequent routing queries (see Appendix A for geometric details).

Accordingly, we represent a pipeline as an ordered sequence of grid points that connects a source s to a target t . The Cartesian coordinates of these points serve as the decision variables of the routing problem: $\text{path} := (p_1, \dots, p_m), p_1 = s, p_m = t, p_i \in \mathcal{X}$, where $\mathcal{X} \subset \mathbb{R}^3$ denotes the discrete set of feasible grid points within the scene bounds at spacing h that are free in the occupancy grid S (i.e., $S[i, j, k] = 1$).

To reconcile the strict engineering codes of petrochemical piping with contemporary sustainability targets, we augment the ant-colony transition rule with an energy-aware surrogate $E_{\text{loc}}(c, n)$. It estimates the incremental energy cost of moving from the current grid node c to a candidate neighbour n . On this basis, we define a multi-factor enhanced heuristic

$$\eta(c, p, n; g) = \frac{1 + \omega_{\text{dir}} R_{\text{dir}}(c, n; g)}{1 + D_{\text{goal}}(n; g)} \cdot \frac{1}{1 + \omega_{\text{h}} P_{\text{h}}(c, n; g)} \cdot \frac{1}{1 + \omega_{\text{bend}} P_{\text{bend}}(p, c, n)} \cdot \frac{1}{1 + \omega_{\text{loc}} E_{\text{loc}}(c, n)} \quad (3.1)$$

with nonnegative weights $\omega_{\text{dir}}, \omega_{\text{bend}}, \omega_{\text{h}}, \omega_{\text{loc}}$ [14, 48]. The heuristic integrates five complementary terms: goal proximity D_{goal} , direction alignment R_{dir} , bend penalty P_{bend} , altitude penalty P_{h} , and the energy-informed surrogate E_{loc} . The geometric components $D_{\text{goal}}, R_{\text{dir}}, P_{\text{bend}}$, and P_{h} encourage deferred elevation, long straight runs, and monotone progress to the target; full expressions are provided in Appendix B.

Energy-informed surrogate term E_{loc} . In petrochemical piping systems, operating energy and carbon emissions are dominated by wall friction, local losses at fittings, and the pumping power required for vertical lifting. To bias the search away from locally energy-intensive moves, we introduce an energy-informed surrogate $E_{\text{loc}}(c, n)$ that approximates the incremental energy cost of moving from the current node c to a candidate node n :

$$E_{\text{loc}}(c, n) = \omega_{\text{friction}} e_{\text{friction}}(c, n) + \omega_{\text{bend}}^E e_{\text{bend}}(p, c, n) + \omega_{\text{elev}} e_{\text{elev}}(c, n), \quad (3.2)$$

where ω_{friction} , ω_{bend}^E , and ω_{elev} are nonnegative coefficients and e_{friction} , e_{bend} , and e_{elev} quantify frictional loss, elbow-induced local loss, and lifting effort, respectively (see Appendix C for explicit forms). Embedded in the denominator of Eq. (3.1), larger values of $E_{\text{loc}}(c, n)$ decrease the local heuristic and steer the ants toward routes that are not only geometrically favourable but also energy- and carbon-efficient under the unified scalar objective.

To handle strict feasibility constraints and evolving free space in industrial pipe routing, we adopt an ACO scheme tailored to this setting. At each step, the ant moves from the current grid node c to a feasible 6-connected neighbor $n \in \mathcal{N}_6(c)$ with probability $\mathbb{P}(c \rightarrow n)$ (see Appendix D for details).

In the path construction stage, we further introduce an adaptive greedy scheduling mechanism: at each expansion step, the next node is selected by a pseudo-random proportional rule with an iteration-dependent [24] greedy threshold $q_0^{(k)}$ [35, 44] that follows a smooth schedule from broad exploration to focused exploitation (see Appendix D for details). This time-varying greediness, combined with directional pheromone deposition and self-regulating updates, mitigates premature convergence and improves route quality under strong process and manufacturability constraints.

3.2 Heterogeneous and Self-Regulating Pheromone Deposition Mechanisms

In classical ACO and in natural foraging, pheromone is deposited almost uniformly along the entire path, supporting bidirectional navigation between nest and food source. In directional engineering tasks such as pipe routing in petrochemical plants, however, the path is inherently one-way from a specified source interface to a unique destination. Under this setting, uniform deposition provides little directional bias, encourages lingering near the source, and promotes convergence to suboptimal detours.

We therefore employ a direction-aware, fitness-sensitive pheromone scheme. First, global evaporation is applied to all path segments,

$$\tau_{ijk}(t+1) \leftarrow (1 - \rho) \tau_{ijk}(t), \quad (3.3)$$

with evaporation rate $\rho \in (0, 1)$ to suppress long-term accumulation on outdated trails and maintain exploration. Next, each constructed path receives a self-regulating reinforcement $\Delta\tau(J)$ that depends on its scalar fitness J , which aggregates energy, environmental impact, length, bends, elevation and clearance violations into a single cost (see Appendix E for details). Higher-quality paths (lower J) are thus reinforced more strongly, sharpening the preference for good solutions.

Finally, reinforcement is modulated along the path by a directional weight $\gamma_{\text{step}} = \delta_{\gamma}^{\frac{S - \text{step}}{\max(S, 1)}}$, where $\delta_{\gamma} \in (0, 1)$ and $\text{step} \in [0, S]$

indexes the position along a path of length S . This yields an update of the form

$$\tau_{ijk}(t+1) = (1 - \rho) \tau_{ijk}(t) + \gamma_{\text{step}} \Delta\tau(J), \quad (3.4)$$

which creates a unimodal pheromone gradient from source to sink and reduces stagnation near the origin. Together, these components accelerate convergence, improve discrimination between competing routes, and mitigate entrapment in suboptimal loops.

3.3 RL Agent Architecture

In 3D pipeline routing for petrochemical plants, process layouts and connection tasks vary markedly across operating scenarios in terms of equipment configurations, process chains, fluid properties, and source–target requirements, making DEACO highly sensitive to its heuristic and control hyperparameters and causing fixed or manually tuned settings to generalize poorly across scenes and connection types. To address this, we introduce GRL-DEACO, in which a PPO-trained agent outputs connection-specific ACO hyperparameters that adapt pheromone emphasis, heuristic balance, and penalty strengths to the current layout and routing requirements.

3.3.1 Actor–Critic architecture with graph-aware perception. We adopt an Actor–Critic reinforcement learning architecture with a Gaussian policy [32]. At decision step t , the agent observes a continuous state $\mathbf{s}_t \in \mathbb{R}^{d_s}$ and samples an action $\mathbf{a}_t \in \mathbb{R}^{d_a}$, which is deterministically decoded into the full DEACO hyperparameter vector for the current connection. In our implementation, the raw connection descriptor has dimension $d_{\text{conn}} = 26$, the action (hyperparameter) vector $d_a = 29$, and the fused state dimension d_s depends on the chosen encoder. The Actor network parameterizes a stochastic policy $\pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t)$ over ACO hyperparameters, while the Critic $V_{\phi}(\mathbf{s}_t)$ provides a state-value baseline for low-variance advantage estimates. As outlined in the problem formulation, at step t the environment is given by an attributed scene graph $\mathcal{G}_t = (\mathcal{V}_t, \mathcal{E}_t, \mathbf{X}_t, \mathbf{A}_t)$ and a current terminal pair (u_t, v_t) . In implementation, $\mathcal{G}_t$ and the connection descriptors are encoded by the graph and connection encoders into a continuous state vector $\mathbf{s}_t$, so that the abstract mapping $\pi : (\mathbf{s}_t, \mathcal{G}_t) \mapsto \psi_t$ is instantiated as $\pi_{\theta} : \mathbf{s}_t \mapsto \mathbf{a}_t$. The learning objective is to maximize the discounted return induced by the downstream ant colony optimizer while remaining robust to heterogeneous 3D petrochemical layouts.

3.3.2 Graph attention encoder. Each device node v is associated with a 24-dimensional feature vector $\mathbf{x}_v \in \mathbb{R}^{24}$ concatenating its 3D position, bounding-box size, port statistics, device-type one-hot encoding, and a dynamic connection-ratio indicator describing how much of the device has been routed in the current episode. Starting from $\mathbf{h}_v^{(0)} = \mathbf{x}_v$, we process the scene graph with a two-layer multi-head GAT [42]. For head $k = 1, \dots, H$ at layer ℓ , we compute

$$\mathbf{z}_v^{(\ell, k)} = \mathbf{W}^{(\ell, k)} \mathbf{h}_v^{(\ell)}. \quad (3.5)$$

$$e_{vu}^{(\ell, k)} = \text{LeakyReLU}\left(\left(\mathbf{a}^{(\ell, k)}\right)^{\top} \left[\mathbf{z}_v^{(\ell, k)} \parallel \mathbf{z}_u^{(\ell, k)}\right]\right), \quad u \in \mathcal{N}(v). \quad (3.6)$$

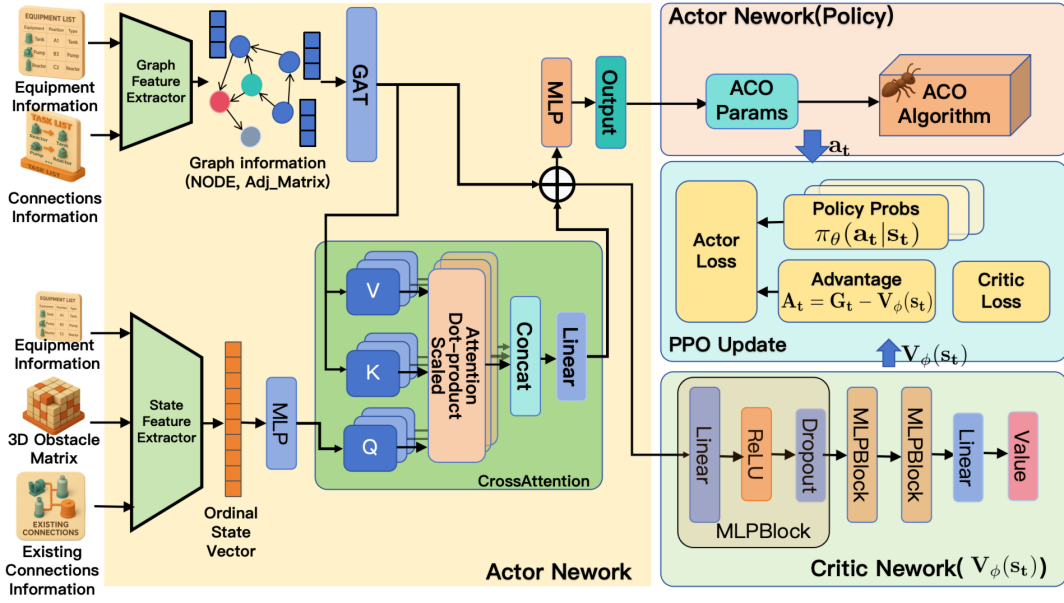


Figure 3: GRL-DEACO: Actor-Critic Architecture with Graph-Aware Perception

$$\alpha_{vu}^{(\ell,k)} = \frac{\exp(e_{vu}^{(\ell,k)})}{\sum_{w \in N(v)} \exp(e_{vw}^{(\ell,k)})}. \quad (3.7)$$

$$\tilde{\mathbf{h}}_v^{(\ell+1,k)} = \sum_{u \in N(v)} \alpha_{vu}^{(\ell,k)} \mathbf{z}_u^{(\ell,k)}. \quad (3.8)$$

$$\mathbf{h}_v^{(\ell+1)} = \sigma\left(\bigoplus_{k=1}^H \tilde{\mathbf{h}}_v^{(\ell+1,k)}\right). \quad (3.9)$$

where $N(v)$ is the neighborhood of v , and $\mathbf{W}^{(\ell,k)}$, $\mathbf{a}^{(\ell,k)}$ are head-specific learnable parameters. After the final layer, we obtain node embeddings $\{\mathbf{h}_v^{(L)}\}_{v \in \mathcal{V}_t}$ and derive a graph-level representation via symmetric pooling

$$\mathbf{g}_t = \text{Pool}(\{\mathbf{h}_v^{(L)} : v \in \mathcal{V}_t\}) \in \mathbb{R}^{d_g}, \quad (3.10)$$

where $\text{Pool}(\cdot)$ combines mean and max pooling to capture both average context and salient bottlenecks in the layout.

Connection encoder and cross-attention fusion. In parallel, the current routing task is encoded as a connection state $\mathbf{x}_t^{\text{conn}} \in \mathbb{R}^{d_{\text{conn}}}$ that fuses: (i) scene-level attributes (normalized volume, obstacle density, and upper bounds on path length, vertical excursion, and clearance), (ii) connection geometry (normalized start/end coordinates, Manhattan and Euclidean distances, and their ratio), (iii) directional cues (dominant direction components and vertical displacement indicators), and (iv) progress statistics (number of completed connections and normalized routing progress). A lightweight MLP encoder

$$\mathbf{c}_t = f_{\text{conn}}(\mathbf{x}_t^{\text{conn}}) \in \mathbb{R}^{d_c} \quad (3.11)$$

maps this descriptor into a dense embedding compatible with the graph representation.

To focus on devices most relevant to the current connection, we introduce a cross-attention fusion module. Treating $\mathbf{c}_t$ as query and node embeddings $\{\mathbf{h}_v^{(L)}\}_{v \in \mathcal{V}_t}$ as keys and values, we form

$$\mathbf{q}_t = \mathbf{W}_q \mathbf{c}_t, \quad \mathbf{k}_v = \mathbf{W}_k \mathbf{h}_v^{(L)}, \quad \mathbf{v}_v = \mathbf{W}_v \mathbf{h}_v^{(L)}, \quad (3.12)$$

with learnable projection matrices $\mathbf{W}_q \in \mathbb{R}^{d_k \times d_c}$ and $\mathbf{W}_k, \mathbf{W}_v \in \mathbb{R}^{d_k \times d_h}$. Attention weights over devices are

$$\alpha_{t,v} = \frac{\exp(\mathbf{q}_t^\top \mathbf{k}_v / \sqrt{d_k})}{\sum_{u \in \mathcal{V}_t} \exp(\mathbf{q}_t^\top \mathbf{k}_u / \sqrt{d_k})}, \quad v \in \mathcal{V}_t, \quad (3.13)$$

and the attended graph context is

$$\tilde{\mathbf{g}}_t = \sum_{v \in \mathcal{V}_t} \alpha_{t,v} \mathbf{v}_v \in \mathbb{R}^{d_k}. \quad (3.14)$$

In practice, we employ a multi-head variant of this cross-attention and aggregate the head outputs, but present the single-head formulation for clarity. The final state representation shared by the Actor and Critic is obtained by concatenation

$$\mathbf{s}_t = [\mathbf{g}_t \parallel \tilde{\mathbf{g}}_t \parallel \mathbf{c}_t] \in \mathbb{R}^{d_s}, \quad (3.15)$$

which jointly encodes global layout structure, connection-specific context, and their interaction.

Actor network and Gaussian policy. On top of the encoded state $\mathbf{s}_t \in \mathbb{R}^{d_s}$, we model the policy as a multivariate diagonal Gaussian over a d_a -dimensional continuous action vector,

$$\pi_\theta(\mathbf{a}_t | \mathbf{s}_t) = \mathcal{N}\left(\boldsymbol{\mu}_\theta(\mathbf{s}_t), \text{diag}(\boldsymbol{\sigma}_\theta^2(\mathbf{s}_t))\right), \quad (3.16)$$

where $\boldsymbol{\mu}_\theta, \boldsymbol{\sigma}_\theta : \mathbb{R}^{d_s} \rightarrow \mathbb{R}^{d_a}$ are output by the Actor network with parameters θ , and d_a equals the number of routing hyperparameters being controlled.

Concretely, we first sample an *unconstrained* action $\tilde{\mathbf{a}}_t \sim \mathcal{N}(\boldsymbol{\mu}_\theta(\mathbf{s}_t), \text{diag}(\sigma_\theta^2(\mathbf{s}_t)))$, which serves as an intermediate latent control in $\mathbb{R}^{d_a}$, and then deterministically transform it into a valid hyperparameter configuration for the ant-colony-based router.

To respect the physical and numerical constraints of the underlying solver, $\tilde{\mathbf{a}}_t$ is first squashed and then linearly mapped into the admissible bounds of each hyperparameter. Let

$$\hat{\mathbf{a}}_t = \frac{1}{2}(\tanh(\tilde{\mathbf{a}}_t) + \mathbf{1}) \in [0, 1]^{d_a}, \quad (3.17)$$

where $\mathbf{1} \in \mathbb{R}^{d_a}$ denotes the all-ones vector, and let $[p_i^{\min}, p_i^{\max}]$ denote the prescribed range of the i -th hyperparameter (e.g., pheromone evaporation rate, heuristic exponents, anisotropic vertical weights, green-objective weights, and algorithmic budgets). The final constrained action $\mathbf{p}_t = (p_{t,1}, \dots, p_{t,d_a})^\top \in \mathbb{R}^{d_a}$ is given componentwise by $p_{t,i} = p_i^{\min} + \hat{a}_{t,i}(p_i^{\max} - p_i^{\min})$, $i = 1, \dots, d_a$, so that each $p_{t,i}$ is the concrete value of the i -th routing hyperparameter at time t . A lightweight validation layer further clips or repairs extreme combinations to guarantee that the downstream ant-colony optimizer remains numerically stable. The magnitude of such corrections is recorded and later incorporated into the reward, so as to discourage pathological predictions.

Critic network and value baseline. At decision step t , the environment emits a scalar reward $r_t = r(\mathbf{s}_t, \mathbf{a}_t)$ given state $\mathbf{s}_t$ and action $\mathbf{a}_t$. The Critic branch shares the same input $\mathbf{s}_t$ as the Actor, but uses an independent MLP of identical depth and width, parameterized by ϕ , to approximate the state-value function

$$V_\phi(\mathbf{s}_t) = f_V(\mathbf{s}_t) \approx \mathbb{E} \left[\sum_{k=0}^{\infty} \gamma^k r_{t+k} \mid \mathbf{s}_t \right], \quad (3.18)$$

with discount factor $\gamma \in (0, 1)$.

Given a sampled trajectory $\{(\mathbf{s}_t, \mathbf{a}_t, r_t)\}_{t=0}^{T-1}$, we compute the empirical discounted return

$$G_t = \sum_{k=0}^{T-t-1} \gamma^k r_{t+k}, \quad (3.19)$$

which may be normalized or softly rescaled according to its empirical mean and variance within a batch to improve numerical stability. The corresponding advantage estimate is $A_t = G_t - V_\phi(\mathbf{s}_t)$, providing a state-dependent baseline that reduces the variance of policy gradient updates.

PPO objective and optimization. Policy optimization follows the clipped PPO objective. Let π_{old} denote the behaviour policy used during data collection. For each transition $(\mathbf{s}_t, \mathbf{a}_t)$, we define the importance-sampling ratio

$$\rho_t(\theta) = \frac{\pi_\theta(\mathbf{a}_t \mid \mathbf{s}_t)}{\pi_{\text{old}}(\mathbf{a}_t \mid \mathbf{s}_t)}. \quad (3.20)$$

The actor loss is the standard clipped surrogate with entropy regularization,

$$\begin{aligned} \mathcal{L}_{\text{actor}}(\theta) = \mathbb{E}_t \left[-\min \left(\rho_t(\theta) A_t, \text{clip}(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon) A_t \right) \right. \\ \left. - \beta_{\text{ent}} \mathbb{E}_t [\mathcal{H}(\pi_\theta(\cdot \mid \mathbf{s}_t))] \right], \end{aligned} \quad (3.21)$$

where $\epsilon > 0$ is the clipping parameter, $\beta_{\text{ent}} \geq 0$ controls entropy regularization, and

$$\mathcal{H}(\pi_\theta(\cdot \mid \mathbf{s}_t)) = -\mathbb{E}_{\mathbf{a} \sim \pi_\theta(\cdot \mid \mathbf{s}_t)} [\log \pi_\theta(\mathbf{a} \mid \mathbf{s}_t)] = \frac{1}{2} \sum_{i=1}^{d_a} \log(2\pi e \sigma_{\theta,i}^2(\mathbf{s}_t)) \quad (3.22)$$

is the differential entropy of the Gaussian policy at state $\mathbf{s}_t$.

The critical parameters ϕ are learned by minimizing the mean-squared error between predicted values and returns,

$$\mathcal{L}_{\text{critic}}(\phi) = \mathbb{E}_t \left[(V_\phi(\mathbf{s}_t) - G_t)^2 \right]. \quad (3.23)$$

Actor and critic are optimized separately with Adam[22] and gradient-norm clipping, and each batch of collected rollouts is reused for several PPO epochs. This training procedure yields a stable hyperparameter policy that can adapt the ant-colony router across heterogeneous 3D routing scenarios.

4 Experiments

We construct a 3D petrochemical pipe-routing dataset by procedurally generating synthetic plant scenarios. For each scene, we sample footprint size and platform elevations within prescribed bounds, instantiate equipment of various sizes and types from a component library, and place them using several complementary layout strategies, with repeated large/medium/small units to mimic grouped and row-wise patterns observed in practice. Scenarios are labeled as simple, medium, or complex according to unit count and connection-graph density; process connections are then automatically generated between units, yielding 100 layouts with on average 8–20 pipelines per scene and roughly 1,290 individual routing tasks in total. We adopt a scenario-level split to prevent leakage of connection tasks across subsets: scenarios are stratified by complexity (simple, medium, complex) and then assigned to training, validation, and test sets in an approximate 16:4:5 ratio, yielding 64, 16, and 20 scenarios, respectively.

PPO is trained by cyclically sampling episodes over a fixed set of 64 training layouts, where each episode performs a full routing pass over all connections in the selected scene, repeatedly exploiting limited but structurally diverse data. We adopt conservative hyperparameters ($\gamma = 0.99$, $\lambda_{\text{GAE}} = 0.95$) and apply gradient-norm clipping (0.5/0.3 for actor/critic) to improve stability and sample efficiency.

4.1 Reinforcement Learning Environment

To assess whether ACO provides a suitable reinforcement learning environment, we quantified its stochastic variability under fixed geometry and hyperparameters. In a representative complex layout, several process connections were selected, and for each connection the normalized-fitness ACO solver was executed independently 30 times under identical conditions; we recorded the fitness J for each run and computed the mean, standard deviation, and coefficient of variation (CV) [1]. As shown in Fig. 5, all connections exhibit CV values well below 0.15: some are effectively deterministic, with perfectly overlapping fitness trajectories, and the remainder lie in a narrow band between approximately 0.003 and 0.074, with no signs of instability or divergence. These nearly flat curves and short error bars indicate that, for a fixed scenario and parameter setting,

Table 1: Routing performance on simple, medium, and complex scenarios (mean $\pm$ SD per connection). Simple/Medium/Complex comprise 6/10/4 scenes with 38/95/51 target connections. All DEACO variants and GRL-DEACO (ours) succeed on 100% of targets, whereas GA succeeds on 30/38, 75/95, and 41/51, indicating higher failure risk. Bold denotes the best (lowest) value per scenario and metric.

Scenario (conn)	Method	L [m]	N_{bend}	H_{alt}	E_{op} [kWh/y]	CO_2^{emb} [kg]	CO_2^{op} [kg/y]
Simple (38)	A*	29.34 $\pm$ 15.74	6.37 $\pm$ 4.64	0.91 $\pm$ 1.21	5692.68 $\pm$ 3891.04	159.55 $\pm$ 83.07	2903.27 $\pm$ 1984.43
	GA (no green)	29.20 $\pm$ 15.21	18.50 $\pm$ 4.30	0.76 $\pm$ 0.72	6544.34 $\pm$ 2795.22	184.68 $\pm$ 81.90	3337.62 $\pm$ 1425.56
	DEACO (no green)	30.99 $\pm$ 15.59	7.37 $\pm$ 2.59	0.76 $\pm$ 0.70	4782.30 $\pm$ 2359.12	171.40 $\pm$ 78.96	2438.97 $\pm$ 1203.15
	DEACO (green)	30.60 $\pm$ 15.56	5.79 $\pm$ 1.30	0.76 $\pm$ 0.70	4494.02 $\pm$ 2318.46	166.33 $\pm$ 77.94	2291.95 $\pm$ 1182.41
	GRL-DEACO (ours)	30.55 $\pm$ 15.56	5.50 $\pm$ 1.07	0.76 $\pm$ 0.70	4442.90 $\pm$ 2314.22	165.50 $\pm$ 77.76	2265.88 $\pm$ 1180.25
Medium (95)	A*	27.61 $\pm$ 14.23	5.22 $\pm$ 3.46	1.79 $\pm$ 2.26	8208.13 $\pm$ 7057.06	149.38 $\pm$ 72.89	4186.15 $\pm$ 3599.10
	GA (no green)	28.73 $\pm$ 13.07	19.69 $\pm$ 4.61	0.99 $\pm$ 0.90	7381.94 $\pm$ 3416.69	184.94 $\pm$ 71.26	3764.79 $\pm$ 1742.51
	DEACO (no green)	30.48 $\pm$ 13.57	7.80 $\pm$ 3.66	1.08 $\pm$ 0.94	5746.15 $\pm$ 3237.49	170.03 $\pm$ 70.31	2930.54 $\pm$ 1651.12
	DEACO (green)	30.05 $\pm$ 13.48	5.91 $\pm$ 2.34	1.07 $\pm$ 0.94	5399.03 $\pm$ 3176.31	164.04 $\pm$ 68.14	2753.51 $\pm$ 1619.92
	GRL-DEACO (ours)	30.00 $\pm$ 13.47	5.58 $\pm$ 1.24	1.07 $\pm$ 0.94	5342.37 $\pm$ 3128.47	163.08 $\pm$ 68.13	2724.61 $\pm$ 1595.52
Complex (51)	A*	29.47 $\pm$ 15.07	6.10 $\pm$ 4.19	1.51 $\pm$ 2.15	14443.02 $\pm$ 50741.17	161.44 $\pm$ 75.52	7365.94 $\pm$ 25878.00
	GA (no green)	30.81 $\pm$ 12.81	19.37 $\pm$ 4.39	1.17 $\pm$ 0.85	7923.01 $\pm$ 3244.04	194.43 $\pm$ 70.19	4040.74 $\pm$ 1654.46
	DEACO (no green)	31.08 $\pm$ 14.16	7.06 $\pm$ 2.34	1.17 $\pm$ 0.87	5929.99 $\pm$ 2996.25	171.29 $\pm$ 72.11	3024.29 $\pm$ 1528.09
	DEACO (green)	30.79 $\pm$ 14.14	5.73 $\pm$ 1.25	1.17 $\pm$ 0.87	5690.78 $\pm$ 2953.99	167.20 $\pm$ 71.01	2902.30 $\pm$ 1506.53
	GRL-DEACO (ours)	30.78 $\pm$ 14.15	5.51 $\pm$ 0.87	1.17 $\pm$ 0.87	5656.50 $\pm$ 2967.66	166.79 $\pm$ 71.24	2884.82 $\pm$ 1513.51

Note. For all results, DEACO (green) uses a single fixed hyperparameter set obtained on a randomly selected scenario by combining three-level grid search and uniform random search over the full parameter space, retaining the configuration with the best fitness. Methods marked as green and GRL-DEACO use an energy- and environment-aware heuristic, while A* and the no green variants do not.

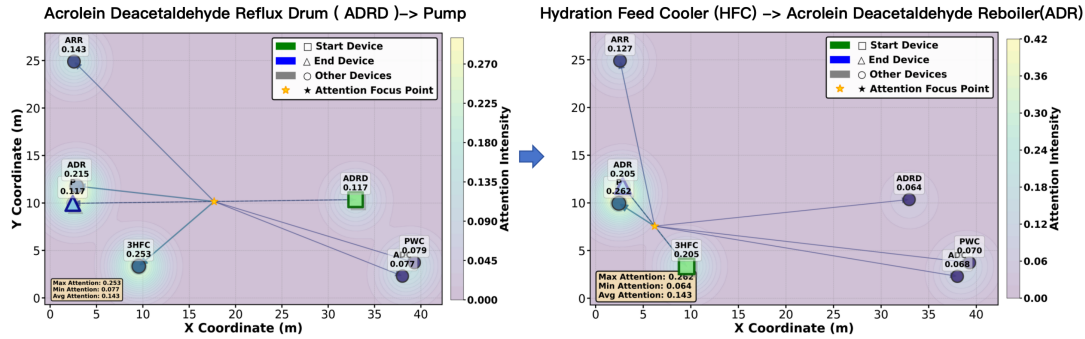


Figure 4: Cross-attention over the 2D layout for two consecutive routing tasks. Left: connection from the Acrolein Deacetaldehyde Reflux Drum (ADRD) to the pump. Right: connection from the Hydration Feed Cooler (HFC) to the Acrolein Deacetaldehyde Reboiler (ADR). In both cases, the cross-fusion attention not only highlights the current start and end devices, but also assigns elevated weights to HFC and ADR, which lie on plausible routing corridors and participate in subsequent process links.

the enhanced-heuristic ACO displays small, bounded, and well-behaved stochasticity in the normalized-fitness space, satisfying the basic stability and learnability requirements for reinforcement-learning-based policy optimization.

4.2 Comparisons

Inspection of Table 1 shows that the direction-enhanced A* baseline attains the smallest mean path length L in medium and complex scenes, while GA is marginally shorter in simple scenes but fails on a subset of target connections. In contrast, A*, all DEACO variants, and GRL-DEACO successfully route every connection at all three

difficulty levels. All constructive methods are evaluated under a single representative operating condition with identical physical parameters (pipe diameter, flow rate, fluid properties, carbon factors; see Appendix F). Non-green variants use the same heuristic weights for length, bends, and elevation penalties, whereas DEACO (green) and GRL-DEACO add energy- and carbon-aware penalty terms on top of DEACO. GRL-DEACO consistently reduces the average number of bends N_{bend} across all layout complexities on the test set, outperforming both A* and fixed-parameter DEACO. Compared with the direction-enhanced A*, the latter often produces elevated routes along high-level corridors, which is undesirable for construction and maintenance (see the right image of Fig. 6). Relative to

Table 2: Mean $\pm$ SD per connection on the test subset with the largest GRL-DEACO-DEACO gap (green), grouped by layout complexity: simple/medium/complex (5/9/2 scenarios; 7/24/8 connections).

Difficulty	Metric	DEACO (green)	GRL-DEACO
Simple	N_{bend}	7.29 ± 1.80	5.71 ± 1.38
	E_{op} [kWh/y]	4390.99 ± 824.29	4159.31 ± 780.92
	CO_2^{op} [kg/y]	2239.40 ± 420.39	2121.25 ± 398.27
Medium	N_{bend}	7.58 ± 4.14	5.83 ± 1.24
	E_{op} [kWh/y]	5633.01 ± 3351.67	5286.94 ± 3295.49
	CO_2^{op} [kg/y]	2872.83 ± 1709.35	2696.34 ± 1680.70
Complex	N_{bend}	7.75 ± 1.83	6.00 ± 1.20
	E_{op} [kWh/y]	4145.14 ± 591.17	3902.96 ± 615.58
	CO_2^{op} [kg/y]	2114.02 ± 301.49	1990.51 ± 313.95

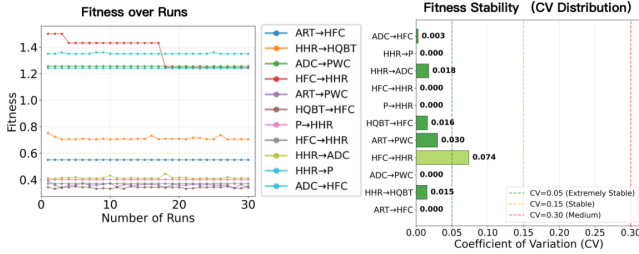


Figure 5: Fitness stability of the ACO environment. Left: normalized fitness over 30 repeated runs per connection with fixed geometry and hyperparameters. Right: CV across connections (mostly $\text{CV} < 0.05$), indicating limited stochasticity and stable RL training.

Table 3: GRL-DEACO inference overhead over 184 routing connections (mean $\pm$ SD). Inference share is the percentage of total time (inference + DEACO planning).

Statistic	Inference (s)	DEACO (s)	Inference / Total (%)
Mean $\pm$ Std	0.6074 ± 0.2842	93.3647 ± 267.2071	1.89 ± 2.12

fixed-parameter DEACO, DEACO (green) mitigates high-elevation detours on some segments; however, for certain connections it still yields elevated sections and excessive local turning (see the left image of Fig. 6 and Table 2). In medium and complex layouts, its operating energy E_{op} and operational emissions CO_2^{op} are systematically lower than those of fixed-parameter DEACO. Even without reinforcement learning, adding the green term to DEACO reduces N_{bend} by about 20% and slightly improves E_{op} , CO_2^{emb} , and CO_2^{op} . GRL-DEACO further amplifies these gains by adaptively balancing geometric and energy-related terms for challenging connection tasks. Table 3 shows that policy inference is lightweight, taking 0.607 ± 0.284 s per connection, while the runtime is dominated by DEACO planning at 93.365 ± 267.207 s.

Fig. 4 further indicates that the cross-fusion attention leverages both the active connection and the scene topology encoded by the

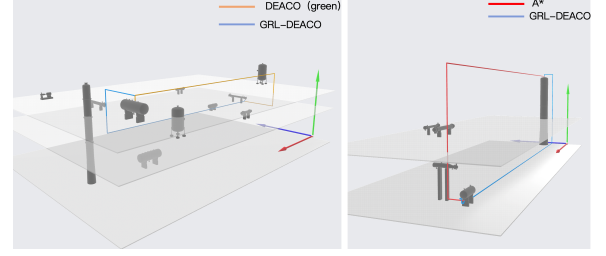


Figure 6: A comparison across scenarios shows distinct elevation behaviors. Fixed-parameter DEACO (green) still produces elevated segments for some connections, whereas A* (red) tends to raise pipes early and route them along high-level corridors. In contrast, GRL-DEACO (blue) postpones elevation and keeps routes closer to the work platform, yielding paths that better align with construction practice.

GAT. Beyond persistently attending to the current source and target devices, it also assigns non-negligible weights to neighbouring equipment located along plausible corridors for upcoming connections. When the source-target pair changes between consecutive tasks, the attention distribution adapts smoothly by shifting its focus to the newly relevant devices. These observations suggest that GRL-DEACO does not solve each connection in isolation; instead, it learns layout-level priors and attends to devices that are topologically and functionally critical for the overall routing plan.

5 Conclusions

In this work, we present GRL-DEACO, an RL-based 3D pipe-routing framework that encodes plant topology and adaptively tunes the hyperparameters of a DEACO solver to improve routing efficiency and solution quality in petrochemical layout design. Energy and environmental indicators are incorporated into the routing objective, and a normalized fitness measure is defined to evaluate DEACO path quality under fixed hyperparameters, providing a stable training signal for the RL agent. A GAT-based encoder with cross-attention captures the equipment and pipe-rack topology, jointly represents the scene state and connection task, and outputs environment-aware ACO hyperparameters.

Experiments on randomized layouts and representative connections show that GRL-DEACO consistently generates 3D routes with fewer bends, improved energy and environmental metrics, and a markedly lower proportion of unnecessary high-elevation segments, while maintaining strong generalization across diverse configurations. Future work will enrich the framework with fine-grained process and safety constraints, incorporate fluid and reaction characteristics into the reward and tuning strategy, and extend GRL-DEACO into a multi-algorithm tuner capable of selecting and optimizing among ACO, PSO, and GA for broader industrial layout and path-planning tasks.

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A Geometric Details of the VBB Representation

Within the scene bounds $[x_{\min}, x_{\max}] \times [y_{\min}, y_{\max}] \times [z_{\min}, z_{\max}]$, each placed device mesh is voxelized at resolution h and hole-filled to obtain a voxel set $\mathcal{V}$. Connected voxel components are approximated by axis-aligned boxes and merged under geometric-error and fill-ratio constraints, yielding a compact box set $\mathcal{B} = \{B_m = [\mathbf{b}_m^{\min}, \mathbf{b}_m^{\max}]\}_{m=1}^{|\mathcal{B}|}$, $\mathbf{b}_m^{\min}, \mathbf{b}_m^{\max} \in \mathbb{R}^3$, $\mathbf{b}_m^{\min} \leq \mathbf{x} \leq \mathbf{b}_m^{\max}$, with $|\mathcal{B}| \ll |\mathcal{V}|$.

To encode clearance constraints, we apply Minkowski dilation with structuring element $\mathbb{B}_{r_{\text{eff}}}$ and define

$$B_m^{\oplus r_{\text{eff}}} = B_m \oplus \mathbb{B}_{r_{\text{eff}}} = \{\mathbf{x} \in \mathbb{R}^3 \mid d(\mathbf{x}, B_m) \leq r_{\text{eff}}\}, \quad (\text{A.1})$$

where $\oplus$ denotes the Minkowski sum, $\mathbb{B}_{r_{\text{eff}}} = \{\mathbf{x} \in \mathbb{R}^3 : \|\mathbf{x}\|_2 \leq r_{\text{eff}}\}$, and $r_{\text{eff}} = r_{\text{pipe}} + r_{\text{safe}}$.

The Euclidean distance from a point $\mathbf{p}$ to an axis-aligned box $B = [\mathbf{b}^{\min}, \mathbf{b}^{\max}]$ used above is $d(\mathbf{p}, B) = \|\max(0, \max(\mathbf{b}^{\min} - \mathbf{p}, \mathbf{p} - \mathbf{b}^{\max}))\|_2$, where the maxima are taken componentwise and $\|\cdot\|_2$ denotes the Euclidean norm.

Let $\Omega = \bigcup_{m=1}^{|\mathcal{B}|} B_m$, $\Omega^+ = \Omega \oplus \mathbb{B}_{r_{\text{eff}}} = \bigcup_{m=1}^{|\mathcal{B}|} B_m^{\oplus r_{\text{eff}}}$. We construct a 3D occupancy grid $S \in \{0, 1\}^{M \times N \times L}$ with voxel spacing h as a rasterization of Ω^+ , where

$$M = \left\lceil \frac{x_{\max} - x_{\min}}{h} \right\rceil, \quad N = \left\lceil \frac{y_{\max} - y_{\min}}{h} \right\rceil, \quad L = \left\lceil \frac{z_{\max} - z_{\min}}{h} \right\rceil. \quad (\text{A.2})$$

Subsequent obstacle-related geometric queries are evaluated on this grid.

B Geometric Heuristic Components

Goal proximity term D_{goal} . Orthogonal rack routing typically advances horizontally before changing elevation, which reduces scaffolding demand, clash risk, and installation difficulty. To discourage premature elevation, we use an anisotropic Manhattan remainder

$$D_{\text{goal}}(n; g) = d_{xz}(n, g) + \alpha_y(d_{xz}(n, g)) d_y(n, g), \quad (\text{B.1})$$

with

$$d_{xz}(n, g) = |n_x - g_x| + |n_z - g_z|, \quad d_y(n, g) = |n_y - g_y|.$$

The horizontal-distance-dependent weight is

$$\alpha_y(u) = 1 + \kappa_y \sigma\left(\frac{u - \Delta_{xz}}{s}\right), \quad \sigma(t) = \frac{1}{1 + e^{-t}}, \quad (\text{B.2})$$

where κ_y , Δ_{xz} , and s are positive hyperparameters. When the remaining horizontal span u is large, $\alpha_y(u)$ approaches $1 + \kappa_y$ and vertical moves are penalized more strongly; as u shrinks near the goal, $\alpha_y(u)$ tends to 1, allowing the final drop.

Direction reward R_{dir} . To suppress zig-zagging and backtracking and promote monotonic progress toward the goal, we reward alignment between the candidate step and the goal direction using a clipped cosine similarity. For current node c , candidate n , and goal g , let

$$\Delta = n - c, \quad u = g - c, \\ \widehat{\Delta} = \frac{\Delta}{\|\Delta\|_2}, \quad \widehat{u} = \frac{u}{\|u\|_2 + \varepsilon_{\text{dir}}},$$

and define

$$R_{\text{dir}}(c, n; g) = \max\{0, \langle \widehat{\Delta}, \widehat{u} \rangle\}, \quad (\text{B.3})$$

where $\varepsilon_{\text{dir}} > 0$ is a small constant for numerical stability.

Bend penalty P_{bend} . Right-angle elbows add fittings and welds, increase head loss, and create potential inspection and leakage points. We therefore penalize orthogonal turns between consecutive Manhattan steps. Let p be the previous grid point, c the current point, and $n \in \mathcal{N}_6(c)$ a candidate successor, with $v_{\text{prev}} = c - p$, $v_{\text{next}} = n - c$. Orthogonal turns are detected via the inner product and penalized as

$$P_{\text{bend}}(p, c, n) = 1\{\langle v_{\text{prev}}, v_{\text{next}} \rangle \leq \varepsilon_{\perp}\}, \quad (\text{B.4})$$

where $\varepsilon_{\perp} > 0$ is a small tolerance and $1\{\cdot\}$ denotes the indicator function.

Altitude penalty P_h . Elevated routing requires additional supports and safety measures and tends to increase clash risk, especially if elevation occurs too early. We therefore couple vertical moves with the remaining horizontal span and the absolute elevation relative to a design reference:

$$P_h(c, n; g) = 1\{|n_y - c_y| = h\} \alpha_y(d_{xz}(n, g)) \\ + \lambda_{\text{alt}} \left(\frac{\max\{0, n_y - y_{\text{ref}}\}}{H_{\text{scale}}} \right)^{\gamma}. \quad (\text{B.5})$$

Here h is the grid spacing, y_{ref} is the reference elevation such as the rack or platform level, and H_{scale} and γ control normalization and growth rate. The first term penalizes premature vertical motion when the remaining horizontal distance is large, while the second term penalizes sustained routing at high elevation.

C Explicit Forms of Energy-related Terms

Friction, elbow, and elevation terms in E_{loc} . For completeness, we give the explicit forms of the components in Eq. (3.2). The elbow-related term is

$$e_{\text{bend}}(p, c, n) = P_{\text{bend}}(p, c, n) \frac{V^2}{2g}, \quad (\text{C.1})$$

where $P_{\text{bend}}(p, c, n) = 1$ if the step $p \rightarrow c \rightarrow n$ forms a right-angle turn and 0 otherwise, V is a characteristic flow velocity, and g is gravitational acceleration.

The lifting-related term is

$$e_{\text{elev}}(c, n) = \max(0, y_n^{\text{world}} - y_c^{\text{world}}), \quad (\text{C.2})$$

where y_n^{world} and y_c^{world} denote the elevations of n and c in global coordinates; it activates only for upward moves ($n_y > c_y$) and captures the additional pumping head required in multi-platform layouts.

The friction term $e_{\text{friction}}(c, n)$ is computed from the segment length and pipe/flow parameters via a standard head-loss model (Darcy–Weisbach)[5], using the shared physical parameters in Appendix F.

D Details of the Adaptive Greedy ACO Scheme

We consider single-source–single-target pipe routing in petrochemical plants, where strong engineering codes, local congestion, and a large combinatorial state space must be respected. In such orthogonal rack and platform layouts, routes should avoid early elevation and unnecessary elbows while preserving maintenance access, hanger placement, and mandated clearances. Classical ACO typically relies on a fixed q_0 to balance exploration and exploitation, which can lead to premature convergence in early iterations or

under heavy congestion. Moreover, uniform pheromone deposition is not well matched to the inherently unidirectional nature of end-to-end pipe routing. To mitigate these issues, we employ an ACO scheme adapted to industrial piping.

At each construction step, an ant moves from the current grid node c to a feasible 6-connected neighbour $n \in \mathcal{N}_6(c)$ with transition probability

$$\mathbb{P}(c \rightarrow n) = \begin{cases} \frac{(\tau(c, n))^\alpha (\eta(c, p, n; g))^\beta}{\sum_{v \in \mathcal{N}_6^{\text{valid}}(c)} (\tau(c, v))^\alpha (\eta(c, p, v; g))^\beta}, & \text{if } n \in \mathcal{N}_6^{\text{valid}}(c), \\ 0, & \text{otherwise,} \end{cases} \quad (\text{D.1})$$

where p is the previous node, g is the goal, τ denotes pheromone, and $\alpha, \beta > 0$ control the relative influence of pheromone and heuristic information. The feasible neighbourhood $\mathcal{N}_6^{\text{valid}}(c)$ is obtained by filtering the 6-connected stencil with the occupancy grid S , retaining only voxels with $S = 1$.

To dynamically balance exploration and exploitation, we adopt an adaptive greedy schedule in the node-selection step. At each expansion, the ant chooses its successor via a pseudo-random proportional rule: a random variable $q \sim \text{Uniform}(0, 1)$ is compared with a greediness threshold $q_0^{(k)}$ at iteration k , and the next node is selected as

$$\text{next_node} = \begin{cases} \arg \max_{n \in \mathcal{N}_6^{\text{valid}}(c)} \mathbb{P}(c \rightarrow n), & \text{if } q < q_0^{(k)} \text{ (greedy),} \\ \text{RouletteWheel}(\mathbb{P}(c \rightarrow n)), & \text{otherwise (sampling).} \end{cases} \quad (\text{D.2})$$

The threshold $q_0^{(k)}$ follows a smooth sigmoid schedule,

$$q_0^{(k)} = \frac{A_{q_0}}{1 + \exp(-\frac{10}{K}(k - 0.5K))} + B_{q_0}, \quad (\text{D.3})$$

where K is the total number of iterations, A_{q_0} controls the amplitude of the rise, and B_{q_0} sets the baseline. Early iterations, therefore, favour stochastic sampling, while later iterations progressively increase the probability of greedy moves, yielding a gradual transition toward exploitation and reducing the risk of stagnation[7, 25] in congested regions.

E Directional and Self-Regulating Pheromone Deposition

In classical ACO and in natural foraging, pheromone is deposited almost uniformly along the entire path, supporting bidirectional navigation between nest and food source. For directional engineering tasks such as pipe routing, however, paths are inherently one-way from a designated source to a unique destination. Under this asymmetry, uniform deposition offers little directional bias, encourages ants to linger near the source, and can reinforce suboptimal detours. We therefore adopt a heterogeneous, self-regulating deposition scheme that couples path quality with distance to the goal.

At each iteration, pheromone on every segment (i, j, k) decays as

$$\tau_{ijk}^{\text{new}} = (1 - \rho) \tau_{ijk}^{\text{old}}, \quad (\text{E.1})$$

where $\rho \in (0, 1)$ is the evaporation rate. After path construction, a fitness-dependent increment is applied. For a path with scalar

fitness J , the global reinforcement term is

$$\Delta\tau = \frac{Q}{\exp\left(\frac{\text{fitness}}{100}\right) - 1 + \varepsilon}, \quad (\text{E.2})$$

with scaling constant Q and small $\varepsilon > 0$ to avoid division by zero, so that high-quality (low-cost) paths receive disproportionately stronger reinforcement.

The fitness J is implemented as a weighted sum of normalized green and geometric indicators,

$$J = w_{op} E_{op}^{\text{norm}} + w_{emb} CO2_{emb}^{\text{norm}} + w_L L^{\text{norm}} + w_{bend} N_{bend}^{\text{norm}} + w_{alt} \text{alt}^{\text{norm}} + w_{clear} \text{viol}^{\text{norm}} + \omega_{\text{Energy}} f_{\text{Energy}} + \omega_{\text{Install}} f_{\text{Install}} + \omega_{\text{Height}} f_{\text{Height}}, \quad (\text{E.3})$$

where the first group encodes operating energy, embodied carbon, length, bend count, elevation and clearance violations, and the last three terms capture higher-level preferences related to energy, installation, and height.

Combining evaporation, fitness-based reinforcement, and directional weighting, the pheromone update at each visited voxel is

$$\tau_{ijk}(t+1) = (1 - \rho) \tau_{ijk}(t) + \gamma_{\text{step}} \Delta\tau, \quad (\text{E.4})$$

which accelerates convergence, mitigates stagnation near the source, and reduces the risk of being trapped in suboptimal loops.

F Physical Parameters for Fair Comparison

All methods are evaluated under a single representative operating condition, corresponding to a DN150 line with fixed flow rate and shared physical and environmental constants. These parameters are identical for A*, DEACO, GA, and GRL-DEACO in all scenarios and are used to compute head loss, operating energy E_{op} , and embodied and operational CO_2 for each routed path. Specifically,

$$\begin{aligned} D &= 0.154 \text{ m}, & Q &= 0.0278 \text{ m}^3/\text{s}, & h &= 0.10 \text{ m}, \\ \gamma_{\text{pipe}} &= 5.0 \text{ kgCO}_2/\text{m}, & \gamma_{\text{elbow}} &= 2.0 \text{ kgCO}_2, & D_{\text{ref}} &= 0.154 \text{ m}, \\ p_{\text{diam}} &= 2.0, & \kappa_{\text{grid}} &= 0.51 \text{ kgCO}_2/\text{kWh}, \\ f &= 0.02, & \rho &= 1000 \text{ kg/m}^3, & \eta_{\text{pump}} &= 0.75, \\ g &= 9.81 \text{ m/s}^2, & T_{\text{year}} &= 8000 \text{ h/yr}. \end{aligned}$$

G Reward design.

For successful routing we use a bounded weighted sum of six pre-normalized terms

$$r_t^{\text{succ}} = 0.35 \tilde{r}_{\text{fit}} + 0.20 \tilde{r}_{\text{len}} + 0.20 \tilde{r}_{\text{h}} + 0.10 \tilde{r}_{\text{corr}} + 0.10 \tilde{r}_{\text{green}} + 0.05 \tilde{r}_{\text{stab}}, \quad (\text{G.1})$$

where the base fitness is normalized online by EMA-based Z-score and a tanh squashing,

$$z = \frac{f - \mu_f}{\sigma_f + 10^{-6}}, \quad \tilde{r}_{\text{fit}} = \text{clip}(-5(1 + \tanh z)), \quad (\text{G.2})$$

and the remaining components $\tilde{r}_{\text{len}}$, $\tilde{r}_{\text{h}}$, $\tilde{r}_{\text{corr}}$, $\tilde{r}_{\text{green}}$, $\tilde{r}_{\text{stab}}$ encode path length, vertical motion, correction ratio, green metrics, and DEACO stability, respectively, each rescaled to a comparable range.